

Disproof of conjecture 00000003844

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Verdict: FALSE

We disprove conjecture 00000003844 as stated. The refutation is unconditional and completely elementary. Let $H(S_3)$ be the 0-Hecke monoid of the finite Coxeter system $(S_3, \{s_1, s_2\})$; it has 6 elements. We exhibit two *proper* submonoids A, B with $A \cup B = H(S_3)$, so its covering number is at most 2; and no single proper submonoid can cover it, so the covering number is exactly 2. Thus the conjecture's first clause ("every finite-type 0-Hecke monoid has infinite covering number") is false. The same argument applies to $H(S_4)$ (24 elements) and, more generally, to every finite-type 0-Hecke monoid of rank ≥ 2 . The direct-product clause fails as well: $\text{cov}(M_1 \times M_2) \leq \text{cov}(M_1) \text{cov}(M_2)$, so $\text{cov}(H(S_3) \times H(S_3)) \leq 4$, not ∞ .

1 The conjecture, quoted verbatim

From `conjectures/00000003844.md`:

English. Definition: The covering number of a monoid is the least number of proper submonoids needed to cover it, infinity if impossible. Conjecture: Every finite-type 0-Hecke monoid and every plactic monoid has infinite covering number, and arbitrary direct products of these two classes still have infinite covering number. (uncoverable direct product closure)

The conjecture is filed in both English and Chinese; the Chinese original is reproduced verbatim (in Unicode) in `README.md`. The two versions are identical in content. The PDF font does not embed CJK glyphs, so the Chinese text is kept in the markdown copy rather than duplicated here.

2 Definitions and conventions

Definition 1 (Covering number). The *covering number* $\text{cov}(M)$ of a monoid M is the least $k \geq 1$ such that $M = S_1 \cup \dots \cup S_k$ for proper submonoids $S_i \subsetneq M$; if no such finite cover exists, $\text{cov}(M) = \infty$. Here "submonoid" always means a subset containing the identity and closed under the product.

Remark 2. A single proper submonoid never covers M , since a proper subset is not the whole of M . Hence a finite covering number is automatically ≥ 2 .

Definition 3 (0-Hecke monoid). Let (W, S) be a finite Coxeter system. The *0-Hecke monoid* $H(W)$ is the monoid with generators π_s ($s \in S$) and relations

$$\pi_s^2 = \pi_s \quad (s \in S), \quad \text{the braid relations of } (W, S),$$

where the braid relation for $s, t \in S$ with $m_{st} < \infty$ is the usual $\pi_s \pi_t \pi_s \cdots = \pi_t \pi_s \pi_t \cdots$ with m_{st} factors on each side. Its elements are indexed by W , and the product is the *Demazure product*

$$\pi_u \pi_v = \pi_{u \star v}, \quad u \star s = \begin{cases} us, & \ell(us) > \ell(u), \\ u, & \ell(us) < \ell(u), \end{cases}$$

extended over a reduced word of v . The product $\star$ is associative and the generators are idempotent, so $H(W)$ is a monoid with identity π_e .

Definition 4 (Length). For $w \in W$, $\ell(w)$ is the Coxeter length, i.e. the number of positive roots sent negative by w , equivalently the length of a reduced word.

Theorem 5 (Length identity). For all $x, y \in W$, $\ell(x \star y) \geq \max\{\ell(x), \ell(y)\}$.

Theorem 5 is verified exhaustively for S_3, S_4, S_5 (all $6^2, 24^2, 120^2$ pairs, 0 violations) by `reproduce.py`. It follows in general from the subword characterisation of the Demazure product: $x \star y$ is the unique maximal-length element of the Bruhat interval $\{z : z \leq x \text{ and } z \leq y\}$ in the Bruhat order on the set of subwords, whence $x \star y \geq x$ and $x \star y \geq y$ in Bruhat order and therefore $\ell(x \star y) \geq \max\{\ell(x), \ell(y)\}$.

3 The counterexample: $H(S_3)$

The Coxeter system is $(S_3, \{s_1, s_2\})$, of type A_2 , of rank 2. Its 0-Hecke monoid has 6 elements, labelled

$$0 = e, \quad 1 = s_2, \quad 2 = s_1, \quad 3 = s_1 s_2, \quad 4 = s_2 s_1, \quad 5 = w_0,$$

where indices are the ones used by `lean4/Main.lean`. The Demazure product is the following 6×6 table (row = left factor, column = right factor):

$\star$	e	s_2	s_1	$s_1 s_2$	$s_2 s_1$	w_0
e	e	s_2	s_1	$s_1 s_2$	$s_2 s_1$	w_0
s_2	s_2	s_2	$s_2 s_1$	w_0	$s_2 s_1$	w_0
s_1	s_1	$s_1 s_2$	s_1	$s_1 s_2$	w_0	w_0
$s_1 s_2$	$s_1 s_2$	$s_1 s_2$	w_0	w_0	w_0	w_0
$s_2 s_1$	$s_2 s_1$	w_0	$s_2 s_1$	w_0	w_0	w_0
w_0	w_0	w_0	w_0	w_0	w_0	w_0

One checks directly that $\star$ is associative (all $6^3 = 216$ triples), that e is a two-sided identity, that $\pi_{s_1}^2 = \pi_{s_1}$ and $\pi_{s_2}^2 = \pi_{s_2}$, and that the braid relation holds:

$$s_1 \star s_2 \star s_1 = s_1 s_2 \star s_1 = w_0 = s_2 s_1 \star s_2 = s_2 \star s_1 \star s_2.$$

3.1 A two-element cover

Set

$$A = \{e, s_2\}, \quad B = H(S_3) \setminus \{s_2\} = \{e, s_1, s_1 s_2, s_2 s_1, w_0\}.$$

Both are submonoids:

- A contains e and is closed because $s_2 \star s_2 = s_2$ (and e is the identity).

- B contains e . For closure, suppose $x, y \in B$ and $x \star y = s_2$. By Theorem 5, $1 = \ell(s_2) = \ell(x \star y) \geq \max\{\ell(x), \ell(y)\}$, so $\ell(x), \ell(y) \leq 1$, i.e. $x, y \in \{e, s_1, s_2\}$. Since $x, y \neq s_2$ by definition of B , we have $x, y \in \{e, s_1\}$, and the table gives $e \star e = e$, $e \star s_1 = s_1 \star e = s_1$, $s_1 \star s_1 = s_1$, none of which is s_2 . Contradiction; hence $x \star y \neq s_2$ and $x \star y \in B$.

Both are *proper*: $|A| = 2 < 6$ and $|B| = 5 < 6$. Finally

$$A \cup B = H(S_3),$$

because every element of $H(S_3)$ is either s_2 (in A) or one of the other five (in B). Therefore $\text{cov}(H(S_3)) \leq 2$.

3.2 One submonoid does not suffice

By Remark 2 a single proper submonoid cannot cover. For completeness, `reproduce.py` enumerates all 64 subsets of the six elements and confirms that the only submonoid equal to all of $H(S_3)$ is $H(S_3)$ itself. Hence

$$\boxed{\text{cov}(H(S_3)) = 2}.$$

The monoid of the counterexample is finite, of finite type, and its covering number is 2, not ∞ . This alone falsifies the conjecture's first clause.

4 $H(S_4)$ and all finite-type 0-Hecke monoids of rank ≥ 2

The same construction works one rank higher. $H(S_4)$ has $|S_4| = 24$ elements. Exhaustive enumeration finds 10577 submonoids (including $H(S_4)$ itself), and the covering number is exactly 2: for every generator s_i ($i = 1, 2, 3$), the pair

$$A = \{e, s_i\}, \quad B = H(S_4) \setminus \{s_i\}$$

is a 2-cover by exactly the argument of Section 4, using Theorem 5.

Theorem 6. *Let (W, S) be a finite Coxeter system of rank $|S| \geq 2$ and let $H(W)$ be its 0-Hecke monoid. Then $\text{cov}(H(W)) = 2$.*

Proof. Fix any simple reflection $s \in S$ and put $A = \{e, s\}$, $B = H(W) \setminus \{s\}$.

A is a submonoid. It contains e , and $s \star s = s$ (idempotence of the generator).

B is a submonoid. It contains e . Let $x, y \in B$ and suppose for contradiction that $x \star y = s$. By Theorem 5, $1 = \ell(s) = \ell(x \star y) \geq \max\{\ell(x), \ell(y)\}$, so $\ell(x), \ell(y) \leq 1$. The elements of length ≤ 1 are e and the simple reflections, so $x, y \in \{e\} \cup S$; since $x, y \neq s$, in fact $x, y \in \{e\} \cup (S \setminus \{s\})$. If $x = e$ then $x \star y = y \neq s$, and if $y = e$ then $x \star y = x \neq s$. Otherwise $x, y \in S \setminus \{s\}$: if $x = y$ then $x \star y = x \neq s$, while if $x \neq y$ then $x \star y$ is a product of two distinct simple reflections, so $\ell(x \star y) = 2 > 1 = \ell(s)$ and again $x \star y \neq s$. Hence $x \star y \neq s$, so $x \star y \in B$. (This step is where rank ≥ 2 is needed for B and A to be *proper*: A is proper because some $t \in S \setminus \{s\}$ exists, and B is proper because $s \notin B$.)

The cover. $A \cup B = H(W)$ is clear, since $s \in A$ and every other element is in B . Both A and B are proper. Hence $\text{cov}(H(W)) \leq 2$, and by Remark 2, $\text{cov}(H(W)) \geq 2$; therefore $\text{cov}(H(W)) = 2$. $\square$

Corollary 7. *The conjecture’s assertion that every finite-type 0-Hecke monoid has infinite covering number is false for the entire class of rank ≥ 2 . Only the degenerate ranks ≤ 1 behave differently: for $W = \{e\}$ (rank 0), $H(W) = \{e\}$ has no proper submonoid, so $\text{cov}(H(W)) = \infty$; for $W = S_2$ (rank 1), $H(W) = \{e, s\}$ has the single proper submonoid $\{e\}$, which does not cover, so again $\text{cov}(H(W)) = \infty$.*

Since the conjecture’s first clause is a universally quantified conjunction over two classes of monoids, the single counterexample $H(S_3)$ already refutes it; no assertion about plactic monoids is needed.

5 The direct-product clause also fails

Proposition 8. *For monoids M_1, M_2 ,*

$$\text{cov}(M_1 \times M_2) \leq \text{cov}(M_1) \cdot \text{cov}(M_2),$$

with the convention $n \cdot \infty = \infty$. Consequently, if both factors have finite covering number, so does the product.

Proof. Let $\text{cov}(M_1) = p$ and $\text{cov}(M_2) = q$ with finite covers by proper submonoids $M_1 = A_1 \cup \dots \cup A_p$ and $M_2 = B_1 \cup \dots \cup B_q$. Then

$$M_1 \times M_2 = (A_1 \cup \dots \cup A_p) \times (B_1 \cup \dots \cup B_q) = \bigcup_{i=1}^p \bigcup_{j=1}^q A_i \times B_j.$$

Each $A_i \times B_j$ is a submonoid of $M_1 \times M_2$: it is closed componentwise and contains (e_1, e_2) . It is proper: since A_i is a proper submonoid of M_1 and $B_j \neq \emptyset$, choose $a \in M_1 \setminus A_i$ and $b \in B_j$; then $(a, b) \notin A_i \times B_j$. Hence the pq proper submonoids $A_i \times B_j$ cover $M_1 \times M_2$, giving the bound. $\square$

Corollary 9. $\text{cov}(H(S_3) \times H(S_3)) \leq 2 \cdot 2 = 4 < \infty$.

Proof. Apply Proposition 8 with $M_1 = M_2 = H(S_3)$ and $\text{cov}(H(S_3)) = 2$ from Section 4. $\square$

`reproduce.py` constructs the four proper submonoids $A \times A$, $A \times B$, $B \times A$, $B \times B$ of the 36-element product $H(S_3) \times H(S_3)$ explicitly (with $A = \{e, s_2\}$ and $B = H(S_3) \setminus \{s_2\}$), verifies that each is closed and proper, and verifies that their union is the whole product. Thus the conjecture’s third clause — that arbitrary direct products still have infinite covering number — is false as well.

Remark 10. The fourth and final ingredient of the conjecture, the claim about plactic monoids, is not needed: refuting either half of the first clause suffices to refute the universal statement. We emphasise that no claim is made here about the covering numbers of plactic monoids; the conjecture is simply false as a whole because of its 0-Hecke half and its product clause.

6 Formalisation

The counterexample is formalised in the Lean 4 project under `lean4/`, with **no imports at all** (no `Std`, no `Mathlib`), no `sorry`, and pinned to `leanprover/lean4:v4.33.1`. Everything is hand-rolled over `List/Nat`, since `Finset` and the `Monoid/Submonoid` typeclasses are unavailable without imports. The main statements are

```

theorem covering_number_is_two :
  sub A = true ∧ sub B = true ∧ A.length < 6 ∧ B.length < 6 ∧
  elems.all (fun x => A.contains x || B.contains x) = true ∧
  (List.range 64).all (fun m => !(coverByOne m)) = true
theorem conjecture_00000003844_false :
  ∃ A B : List Nat, sub A = true ∧ sub B = true ∧
  A.length < 6 ∧ B.length < 6 ∧
  elems.all (fun x => A.contains x || B.contains x) = true

```

Here `sub S = S.contains 0 && S.all (fun a => S.all (fun b => S.contains (mul a b)))` is the hand-rolled submonoid predicate, `A = [0,1]` and `B = [0,2,3,4,5]`, and `mul` is the Demazure product read off an explicit 6×6 table. The monoid axioms (`assoc` over all 216 triples, `identity`, `gen_idem_s1`, `gen_idem_s2`, `braid`), the two submonoid statements, the cover, and the exhaustive absence of a 1-cover over all 64 subsets are all discharged by `decide`. The axiom audit (`Check.lean`) reports only `propext` (and no axioms at all for the cardinality and cover statements); `sorryAx` does not appear. See `lean4/README.md` for the full statement table and scope note.

7 Reproducibility

The exhaustive computations are carried out by `reproduce.py` using the Python 3 standard library only:

```

$ python3 reproduce.py
... (builds  $H(S_n)$  for  $n = 2, 3, 4$ ; checks the monoid axioms,
enumerates all submonoids (16 for  $H(S_3)$ , 10577 for  $H(S_4)$ ),
computes the exact covering number, checks the length identity for
 $S_3, S_4, S_5$ , and checks the product bound on  $H(S_3) \times H(S_3)$ )
PASS: conjecture 00000003844 is FALSE.

```

It exits 0 on PASS and non-zero if any check fails.

LaTeX document:

```
tectonic --outdir build main.tex
```

Lean 4 project:

```

cd lean4
lake build
lake env lean Check.lean

```

Status against the submission rules

Rule 3 requires each submission to contain the LaTeX source, a PDF document, and a Lean 4 project.

- **LaTeX source** — `main.tex`, a standalone article using only `geometry`, `amsmath`, `amssymb`, `amsthm`, `array`, `parskip`, `booktabs`; compiles with `tectonic --outdir build main.tex`.
- **PDF document** — present at `build/main.pdf`.

- **Lean 4 project** — present under `lean4/`, with `lean-toolchain` pinned to `leanprover/lean4:v4.33.1`, `lakefile.toml` (library `Main`), `Main.lean`, `Check.lean`, and a `README.md`. It contains no `import` statement, no `Mathlib` dependency, and no `sorry`; the axiom audit reports no `sorryAx` and only `propext`.